

$-2 \Delta \ln L$

**CMS**  
*Internal*

$r = 1.813$   $+0.602$   
 $-0.313$

— Total uncert.  
- - Freeze TTnorm  
- - Freeze PF  
- - Freeze btag  
- - Freeze Pileup  
- - Freeze MET  
- - Freeze lepton  
- - Freeze all

- - Freeze theory  
- - Freeze OSnorm  
- - Freeze lumi  
- - Freeze jet  
- - Freeze VBS  
- - Freeze tau  
- - Freeze mischarge

$r = 1.813$   $+0.213$   $-0.000$  (theory)  $+0.000$  (TTnorm)  $+0.003$  (OSnorm)  $+0.079$  (PF)  $+0.061$  (lumi)  $+0.000$  (btag)  $+0.002$  (jet)  $+0.137$  (Pileup)  $+0.036$  (VBS)  $+0.060$  (MET)  $+0.053$  (tau)  $+0.000$  (lepton)  $+0.000$  (mischarge)  $+0.183$  (MCstat)  $+0.513$  (Stat)  
 $-0.000$   $-0.000$   $-0.046$   $-0.009$   $-0.031$   $-0.105$   $-0.246$   $-0.039$   $-0.000$   $-0.170$   $-0.017$   $-0.000$   $-0.000$   $-0.000$   $-0.423$

